

Quiz 03 – Chapter 03 – Storage**[50 Marks, 50 Minutes]**

Name: _____

Roll No: _____

Question 1: [1x12 marks] [estimated time 12 minutes]
Supply the missing words to complete the statements.

1. A storage __Medium__ is the hardware where data is actually stored. (Device / Medium)
2. Conventional RAM is __volatile__ so programs and documents held in RAM are erased when they are no longer needed by the computer or when the power to the computer is turned off.
3. Random access, also called __direct access__, means that data can be retrieved directly from any location on the storage medium, in any order.
4. Storage devices used like hard drives, CD/DVD drives, and USB flash drives are __Random Access__ devices. (Random Access / Sequential Access)
5. A newer type of hard drive is the __SSD__ that uses flash memory technology instead of magnetic technology to store data.
6. Read/write heads the magnetize particles to store, retrieve and access data are present in __magnetic hard__ drives.
7. __Track__ is concentric path on a disk where data is recorded. (Track / Register)
8. __Optical Disc__ is a type of storage medium read from and written to using a laser beam.
9. Storage capacity of CD and DVD is __700MB__ and __4.7GB__ respectively. (25GB/5GB, 7GB/4GB, 700MB/4.7GB, 8.5GB/700MB).
10. NAS and SAN stand for __Network Attached Storage__, __Storage Area Network__.
11. __RAID__ is a method of storing data on two or more hard drives that work together. Although it can

be used to increase performance, it is most often used to protect critical data on a large storage system.

12. __Cloud Storage__ refers to using a storage device that is accessed via the Internet.

Question 2: [1x5 marks] [estimated time 3 minutes]
Write the following storage capacities (700MB, 4.7GB, 8.5 GB, 25 GB, 50 GB, 100 GB) on right places.

- a) CD __700MB__.
- b) DVD __4.7GB__.
- c) BD __25GB__.
- d) DVD-DL __8.5GB__.
- e) BD-DL __50GB__.

Question 3: [1x8 marks] [estimated time 8 minutes]
Circle T if the statement is true and F otherwise.

1. **T F** A computer system with a C drive and a D drive must have two physical hard drives
2. **T F** A hybrid hard drive contains both magnetic hard disks and optical discs.
3. **T F** A DVD-RW disc can be written to and rewritten to.
4. **T F** Secure Digital (SD) cards are one type of magnetic storage medium.
5. **T F** Media that allow random access are sometimes referred to as addressable media.
6. **T F** Hard drive has separate storage medium and storage device.
7. **T F** Holographic storage is emerging type of 2D storage technology.
8. **T F** The tiny depressions, dark areas, or otherwise altered spots on an optical disc that are used to represent data are called track.

**Question 4: [5x5 marks] [estimated time 25 minutes]
Give Short answers to the followings**

1. Write comparison of SSD with HDD?

Solid-state drives (SSDs) are hard drives that use flash memory technology instead of spinning hard disk platters and magnetic technology. consequently,

SSDs have no moving parts and data is stored as electrical charges on the flash memory media located within the SSD.

HDD is cheaper than SSD. In addition to cost, another disadvantage of SSDs is that flash memory cells can wear out with repeated use. HDD has spinning platters and magnetic technology. HDD has longer lifespan.

2. List two possible advantages and two possible disadvantages for using cloud storage?

Advantages:

Cost- Purchasing physical storage can be expensive. Without the need for hardware cloud storage is exceptionally cheaper per GB than using external drive

Accessibility- Using the cloud for storage gives you access to your files from anywhere that has an internet connection.

Recovery - In the event of a hard drive failure or other hardware malfunction, you can access your files on the cloud.

Disadvantages:

Internet Connection - Cloud based storage is dependent on having an internet connection.

Privacy -

When you use a cloud provider, your data is no longer on your physical storage. So, who is responsible for making sure that data is secure?

3. Differentiate Disk stripping and mirroring in RAID?

Striping: When a file is written to a RAID system using striping, it is split among two or more drives.

Mirroring: When a file is written to a RAID system using mirroring, an identical copy of the file is sent to another drive in the system.

4. Which types of storage media would be appropriate for someone who needed to exchange large (5 MB to 75 MB) files with another person? List at least three different types, stating why each might be the most appropriate under specific conditions?

Recordable CD (CD-R) disc (if sending individuals had a CD drive; especially appropriate if media will not be returned); Zip disk (if both individuals had a Zip drive, such as exchanges within a single office); flash memory card (if both individuals had a compatible reader and the media can be returned, such as exchanges within a single office or between friends); flash memory drive (if both individuals had a USB drive and the media can be returned, such as exchanges within a single office or between friends).

5. What are the three steps that hard disk takes for a hard drive to read or write data? (Steps of disk access time)

1. Move the read/write heads to the cylinder that contains (or will contain) the desired data—called seek time.

2. Rotate the hard disks into the proper position so that the read/write heads are located over the part of the cylinder to be used—called rotational delay.

3. Move the data, such as reading the data from the hard disk and transferring it to memory, or transferring the data from memory and storing it on the hard disk— called data movement time.